

**EN – 137**

VI Semester B.Tech. (CSE/ISE/AI&ML) Examination, December 2024
(NEP Scheme)
21CCPC 602 : MACHINE LEARNING

Time : 3 Hours

Max. Marks : 100

Instruction : Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.

1. a) Define Machine learning. Discuss with examples various categories of machine learning algorithm and its applications. 10
- b) Discuss concept learning as search with respect to general to specific ordering of hypothesis. Illustrate Find-S algorithm over the enjoy sport training instances given.

Example	Sky	Air temp	Humidity	Wind	Water	Forecast	Enjoy Sport
1	Sunny	Warm	Normal	Strong	Warm	Same	Yes
2	Sunny	Warm	High	Strong	Warm	Same	Yes
3	Rainy	Cold	High	Strong	Warm	Change	No
4	Sunny	Warm	High	Strong	Cool	Change	Yes

10

OR

2. a) Define machine learning. Explain how some areas/disciplines influenced the machine learning concept. 10
- b) Write Candidate elimination algorithm and apply candidate elimination algorithm to obtain the version space for the given training examples.

Eyes	Nose	Head	Color	Hair	Smile
Round	Triangle	Round	Purple	Yes	Yes
Square	Square	Square	Green	Yes	No
Round	Triangle	Round	Yellow	Yes	Yes
Round	Triangle	Round	Green	No	No
Square	Square	Round	Yellow	Yes	Yes

10

P.T.O.



3. a) Explain the concept of entropy and information gain. Describe the ID3 algorithm for decision tree learning. 7
- b) Differentiate between two types of biases. Why prefer short hypotheses ? 5
- c) Calculate entropy and information gain of A1 and A2 for the following dataset and construct decision tree. 8

Instance	Classification	A1	A2
1.	+	T	T
2.	+	T	T
3.	–	T	F
4.	+	F	F
5.	–	F	T
6.	–	F	T

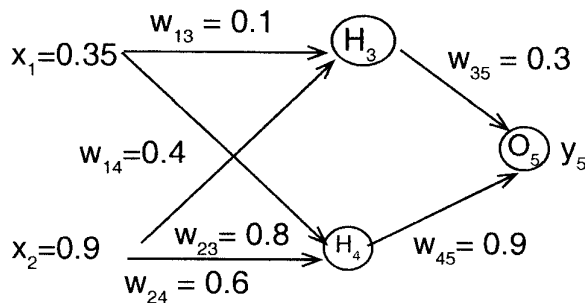
OR

4. a) What are issues in decision tree learning ? Explain briefly how are they overcome ? 10
- b) Construct decision tree using ID3 given the following training examples for the target function play Tennis.

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No



5. a) Discuss Perceptron training rule and Gradient descent training rule. 10
- b) Assume that the neurons have a sigmoid function, perform forward pass and a backward pass on the network. Assume that the actual output of the network is 0.5 and learning rate is 1. Compute new updated weights to the output unit. 10



OR

6. a) What is linearly separable and linearly inseparable problem. Design a network of perceptron to implement X and Y. 10
- b) Write and explain Back Propagation algorithm. Mention its limitations. 10
7. a) Define Bayesian theorem and MAP hypothesis. Consider a medical diagnosis problem in which there are two alternative hypotheses : 1. That the patient has a particular form of cancer (+) and 2. That the patient does not (-). A patient takes a lab test and the result comes back negative. The test returns a correct positive result in only 98% of the cases in which the disease is actually present, and a correct negative result in only 97% of the cases in which the disease is not present. Furthermore, .008 of the entire population have this cancer. Determine whether the patient has cancer or not using MAP hypothesis. 10
- b) Explain Naive Bayes classifier and apply it to the given table above in (4b) for novel instance < outlook = sunny, temp = cool, humidity = high, wing = strong>. 10

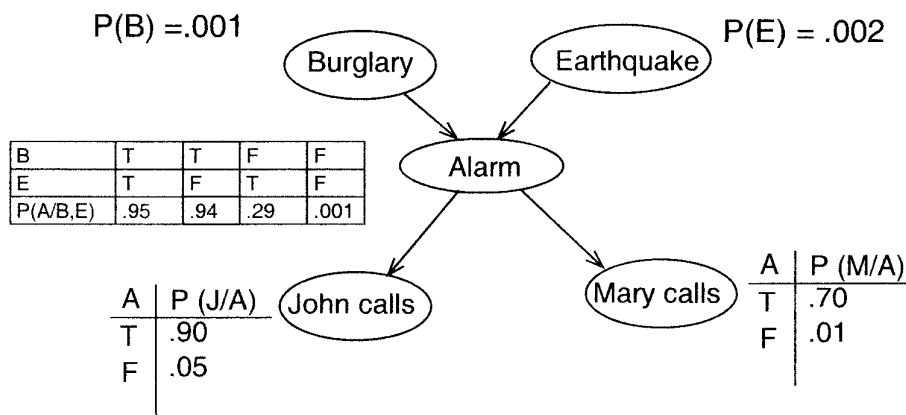
OR

8. a) Define MAP hypothesis. Derive an expression for Maximum Likelihood (ML) and least squared error hypothesis for learning. 10
- b) What is Bayesian Belief Network ? Hari installed a new burglar alarm at his home to detect burglary. The alarm reliably responds at detecting a burglary but also responds for minor earthquakes. Hari has two neighbors Deepak



and Seema, who have taken a responsibility to inform Hari at work when they hear the alarm. Deepak always calls Hari when he hears the alarm, but sometimes he got confused with the phone ringing and calls at that time too. On the other hand, Seema likes to listen to high music, so sometimes she misses to hear the alarm. Represent the probability distribution of events. Calculate the probability that alarm has sounded, but there is neither a burglary, nor an earthquake occurred and Deepak and Seema both called the Hari.

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9. a) What is the use of Principal component analysis in machine learning ?
Discuss with a real life example.

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- b) Bring out the differences between singular value decomposition and matrix factorization. Compute the SVD of the matrix.

10

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

OR

10. a) Explain locally weighted linear regression.

7

- b) Following table gives data of classification of tissue paper if good or bad based on two attributes (acid durability and strength). Classify new tissue paper test with $X_1 = 5$ and $X_2 = 4$ using k-NN algorithm with $k = 3$.

X1	X2	Classification
7	7	Bad
7	4	Bad
3	4	Good
1	4	Good

- c) Explain the reinforcement learning approach and Q learning task.

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